

KEY VALUE PAIRS

HashMap ==> Insertion order is not maintained

LinkedHashMap

TreeMap

HashSet

==> Internally uses Hash Table to store the elements

==> Insertion order is not maintained

==> cannot store duplicates

LinkedHashSet

==> Internally uses Hash Table to store the elements

==> Insertion order is maintained

==> cannot store duplicates

TreeSet

==> Internally uses Hash Table to store the elements

==> Sorting purpose

==> cannot store duplicates

Number Theory Problems

1) check for prime number

factor ==> a factor is a number that divides another no without remainder

32 ==> 1, 2, 4, 8, 16, 32(factors)

48 ==> 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

3 ==> 1, 3

5 ==> 1, 5

7 ==> 1, 7

9 ==> 1, 3, 9

11 ==> 1, 11

3, 5, 7, 11, ==> Prime Number

Given N, determine whether it is prime or not

N ==> 3 ==> P.N

N ==> 36 ==> N.P.N

N ==> 39 ==> N.P.N

N = 41 ==> P.N

N ==> 41 ==> CHECK? true OR false

N = 17

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17

N = 8888

N/1 ==> 0?

.....

N/8888 ==> 0?

we use the use the approach

sqrt(N)

N ==> 25

==> 1, 5, 25

N = 48

1 * 48 ==> 48

2 * 24 ==> 48

3 * 16 ==> 48

4 * 12 ==> 48

6 * 8 ==> 48

8 * 6 ==> 48

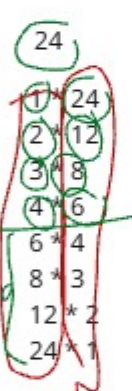
12 * 4 ==> 48

16 * 3 ==> 48

24 * 2 ==> 48

48 * 1 ==> 48

sqrt(48) ==> 6.9 ==> 6



Give N, print its prime factors that multiples the result

N = 60 ==> 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

1, 2, 3, 5 ==> prime factors

2 * 2 * 3 * 5 ==> N(60)

36 ==> 1, 2, 3, 4, 6, 9, 12, 18, 36(factors)

36 ==> (1, 2, 3) prime factors

2 * 2 * 3 * 3 ==> N

36 ==> 4, 2, 3

2, 2, 3, 3

N ==> 36 / 2 ==> 18 ==> 0

18 / 2 ==> 9 ==> 0

9 / 3 ==> 3 ==> 0

3 / 3 ==> 1 ==> 0

48 ==> 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

48 ==> 1, 2, 3

2, 2, 2, 2, 3

48 / 2 ==> 24 ==> 0

24 / 2 ==> 12 ==> 0

12 / 2 ==> 6 ==> 0

6 / 2 ==> 3 ==> 0

3 / 3 ==> 1 ==> 0

Find GCD of 48 and 36 -> 12

48 ==> 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

36 ==> 1, 2, 3, 4, 6, 9, 12, 18, 36

gcd(48, 36)

48 - 36 ==> 12

gcd(12, 36)

36 - 12 ==> 24

gcd(12, 24)

24 - 12 ==> 12

gcd(12, 12)

12 - 12 ==> 0

gcd(48, 36) ==> 12

gcd(48, 36)

48 - 36 ==> 12

gcd(12, 36) ==> 12

36 - 12 ==> 24

gcd(12, 24) ==> 12

24 and 20 ==> 4

24 ==> 1, 2, 3, 4, 6, 8, 12, 24

20 ==> 1, 2, 4, 5, 10, 20

24 - 20 ==> 4

20 and 4 ==> 16

16 and 4 ==> 12

12 and 4 ==> 8

8 and 4 ==> 4

4 and 4